Deakin University

Research Project/Research Project (Advanced)

OnTrack Submission

Renewed Supervision Agreement

Submitted By:
N/a Nikhil
s224234862
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Tutor:
Ananda Maiti

Task Outcomes		Supports
TLO1	Demonstrate engagement with the following unit learning outcomes (legacy)	ULO5 Application of Skills and Knowledge

March 8, 2026

Supervisor Agreement – SIT724/SIT746

Student Name:	Nikhil
Supervisors' Names:	Dr. Ananda Maiti
Project Title:	Detection and Simulation of GPS Spoofing in Mobile Robots Using Sensor Cross-Validation
Is the project continued from SIT723?	Yes
If the project is NOT continued from SIT723, what is the main reason?	NA
Student Details	
Course:	Bachelors of Software Engineering (Honours) Unit: SIT 746 Research Project (Advanced)

Research Experience:	1. Completed SIT723/SIT792 with extensive research on GPS spoofing detection, including a 42-page literature review, methodology development, and vulnerability analysis specific to autonomous mobile robots. 2. Developed a fully containerised 3-container PX4–Gazebo–ROS2 simulation architecture for real-time GPS data capture, sensor streaming, and controlled spoofing scenario preparation. 3. Conducted systematic analysis of sensor fusion techniques (GPS, IMU, odometry), Dynamic Time Warping approaches, and temporal ML-based spoofing detection methods. 4. Implemented real-time GPS coordinate monitoring, pattern analysis, and MAVLink data pipelines as the foundation for algorithm development.
Technical Skills:	**1. Robotics & Simulation:** PX4 SITL, MAVROS, ROS2, Gazebo, jMAVSim, Docker containerisation, real-time sensor streaming, MAVLink protocol. **2. Machine Learning:** LSTM networks, temporal sequence modelling, sensor fusion ML pipelines, anomaly detection modelling, trust index concepts. **3. Software Development:** Python, C++, PyTorch, NumPy, SciPy, scikit-learn, Ubuntu/Linux environments, Docker networking. **4. Cybersecurity (Autonomous Systems):** GPS spoofing taxonomy, threat modelling, vulnerability analysis, signal-level detection concepts, defensive research methodologies. **5. Embedded & Systems:** Real-time system performance optimisation, multi-sensor integration.
Research Units related Information	
SIT724/SIT746 Target Grade:	High Distinction (HD)

Expectations from the unit:	1. Implement and train LSTM-based temporal spoofing detection models using datasets generated from the PX4–Gazebo simulation environment. 2. Develop hybrid sensor fusion detection (GPS–IMU–odometry) using DTW-based cross-validation techniques. 3. Build and validate a machine-learning-based Dynamic GPS Trust Index for continuous reliability assessment. 4. Implement controlled GPS spoofing attack scenarios within the simulation and evaluate detection accuracy, false-positive rates, and computational performance. 5. Produce high-quality research output suitable for publication and complete the SIT746 thesis with strong academic and technical contributions
When will you meet with the supervisor team?	Frequency: Weekly (or bi-weekly, depending on progress). Mode: Remote (Microsoft Teams). Time: Every Thursday 9am.